UiPath Automation Suite 2025.10.2

Multi-Node HA Offline Installation on Linux

Topology: 3 server nodes + 3 agent nodes, air-gapped (offline) environment

Document type: Step-by-step reference runbook, organised node by node

Audience: Platform / infrastructure engineers performing the installation

Important - verify before you run. This runbook was prepared as a reference and has not been validated line-by-line against the official UiPath 2025.10.2 documentation. Command flags, bundle file names and hardware minimums can change between releases. Use it together with the official *Automation Suite on Linux 2025.10 Installation Guide* (docs.uipath.com). If any flag is rejected, run `./bin/uipathctl <command> --help` for the exact syntax of your build.

Example values used throughout (replace with your own):

Item	Example value
Cluster FQDN	automationsuite.corp.local
Load balancer (fixed_rke_address)	automationsuite-lb.corp.local
SQL Server	sql.corp.local:1433
NTP server	ntp.corp.local
Installer directory (all nodes)	/opt/UiPathAutomationSuite/2025.10.2/installer

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1. Execution order at a glance

The installation is **not** done on all nodes at the same time. Follow this order strictly:

Step	Node(s)	Action	Parallel?
1	All 6	OS preparation, copy files, configure disks, copy cluster_config.json, prereq check (Section 4)	Yes
2	Server 1	Install RKE2 infrastructure - creates the cluster (Section 5)	-
3	Server 2	Join as server. Wait until node is Ready (Section 6)	No
4	Server 3	Join as server. Wait until node is Ready (Section 7)	No
5	Agent 1, 2, 3	Join as agents (Sections 8-10)	Yes
6	Server 1	Upload images to in-cluster registry (Section 11)	-
7	Server 1	Install Fabric + UiPath products with manifest apply (Section 11)	-
8	Server 1 / browser	Validate, license, certificates, backup (Section 12)	-

Never join two server nodes at the same time. etcd membership changes must happen one by one. Agent nodes can be joined in parallel once all three servers are Ready.

2. What gets installed on which node

Item	Server 1	Server 2	Server 3	Agent 1-3
OS preparation (NTP, swap off, firewall, packages)	Yes	Yes	Yes	Yes
as-installer.zip (uipathctl, scripts)	Yes	Yes	Yes	Yes
sf-infra.tar.gz (infrastructure offline bundle)	Yes	Yes	Yes	Yes
sf.tar.gz (UiPath products offline bundle)	**Yes**	No	No	No
Optional bundles (DU, CV, ML packages)	If needed	No	No	No
Cluster disk -> /var/lib/rancher	Yes	Yes	Yes	Yes
etcd disk -> /var/lib/rancher/rke2/server/db	Yes	Yes	Yes	No
Data disk -> /datadisk	Yes	Yes	Yes	Optional
Ceph raw disk (in-cluster object store)	Yes	Yes	Yes	Optional
UiPath bundle disk -> /uipath	**Yes**	No	No	No
cluster_config.json (identical copy)	Yes (source)	Yes	Yes	Yes
Kubernetes role	control-plane + etcd	control-plane + etcd	control-plane + etcd	worker
RKE2 service	rke2-server	rke2-server	rke2-server	rke2-agent
uipathctl rke2 install	no -j flag (first node)	-j server	-j server	-j agent
uipathctl rke2 registry upload	**Yes**	No	No	No
uipathctl manifest apply	**Yes**	No	No	No

Node inventory (fill in)

Role	Hostname	IP	vCPU	RAM	Disks (sdb/sdc/sdd/sde/sdf)
Server 1 (primary)	as-srv-01	10.0.0.11	16+	32-64 GB+	
Server 2	as-srv-02	10.0.0.12	16+	32-64 GB+	
Server 3	as-srv-03	10.0.0.13	16+	32-64 GB+	
Agent 1	as-agt-01	10.0.0.21	16+	32-64 GB+	

Role	Hostname	IP	vCPU	RAM	Disks (sdb/sdc/sdd/sde/sdf)
Agent 2	as-agt-02	10.0.0.22	16+	32-64 GB+	
Agent 3	as-agt-03	10.0.0.23	16+	32-64 GB+	

Size every node with the **UiPath Installation Sizing Calculator** for the products you enable. All nodes must run the same supported RHEL version (8.x / 9.x per the 2025.10 compatibility matrix) and be in the same subnet with low latency (< 10 ms between servers).

3. Before you start: infrastructure prerequisites

3.1 Load balancer (L4 / TCP)

Port	Backend nodes	Purpose
443	All nodes	HTTPS - UiPath UI and APIs
6443	Server 1, 2, 3	Kubernetes API
9345	Server 1, 2, 3	RKE2 node registration (join)
80	All nodes (optional)	HTTP -> HTTPS redirect

The LB address becomes `fixed_rke_address` in cluster_config.json. It must be working **before** Server 2/3 and agents join.

3.2 DNS records (point to the LB)

```
automationsuite.corp.local     -> LB VIP
*.automationsuite.corp.local   -> LB VIP   (wildcard)
```

If wildcards are not allowed, create explicit records for the sub-domains listed in the 2025.10 docs (e.g. alm., monitoring., registry., objectstore., insights., apps.).

3.3 Certificates

Server certificate (PEM + key, or PFX) whose SAN covers the FQDN and *.FQDN, plus the full CA chain. Self-signed is acceptable for the first install and can be replaced later.

3.4 External SQL Server

Reachable on port 1433 from **every** node. Login with **dbcreator** (or pre-created databases). Collation SQL_Latin1_General_CP1_CI_AS. Full-Text Search if required by the products you enable.

3.5 Ports between cluster nodes

Port	Proto	Between	Purpose
22	TCP	Admin -> nodes	SSH
80, 443	TCP	Clients / LB -> nodes	Ingress
2379-2380	TCP	Server <-> Server	etcd
6443	TCP	All -> servers / LB	Kubernetes API
9345	TCP	All -> servers / LB	RKE2 supervisor
10250	TCP	All <-> All	kubelet
8472	UDP	All <-> All	Canal / Flannel VXLAN
9099	TCP	All <-> All	Canal health check
9100	TCP	All <-> All	Node exporter
30000-32767	TCP	All <-> All	NodePort range
1433	TCP	Nodes -> SQL	SQL Server

> Simplest approach for a dedicated subnet: allow **all** traffic between the six cluster nodes and restrict only inbound traffic from outside.

3.6 Download the offline files (on an internet-connected machine)

File	Contents	Copy to
as-installer.zip	uipathctl, configureUiPathDisks.sh, installUiPathAS.sh, versions/	All 6 nodes
sf-infra.tar.gz	RKE2, RPMs, infrastructure images	All 6 nodes
sf.tar.gz	UiPath product images and Helm charts	Server 1 only
Optional (du-ondemand, cv-ondemand, ...)	Only for products you enable	Server 1 only

Exact file names are on the 2025.10.2 download page. Verify with `sha256sum` after copying.

```
# from the jump host
for h in as-srv-01 as-srv-02 as-srv-03 as-agt-01 as-agt-02 as-agt-03; do
  scp as-installer.zip sf-infra.tar.gz root@$h:/tmp/
done
scp sf.tar.gz root@as-srv-01:/tmp/
```

4. Common steps for ALL six nodes

Run steps 4.1 - 4.5 as **root** on Server 1, 2, 3 and Agent 1, 2, 3. Only the hostname and the disk command (Step 4.4) differ per node - those are repeated in each node's section.

4.1 Name resolution and time

```
cat >> /etc/hosts <<'EOF'
10.0.0.11 as-srv-01
10.0.0.12 as-srv-02
10.0.0.13 as-srv-03
10.0.0.21 as-agt-01
10.0.0.22 as-agt-02
10.0.0.23 as-agt-03
EOF

dnf install -y chrony            # from local repo / ISO
sed -i 's/^pool .*/server ntp.corp.local iburst/' /etc/chrony.conf
systemctl enable --now chronyd
chronyc tracking                 # offset should be a few ms
```

4.2 Kernel / network settings

```
# swap off
swapoff -a && sed -i '/ swap / s/^/#/' /etc/fstab

# firewall: open ports from Section 3.5, or disable
systemctl disable --now firewalld

# RKE2 requirement
systemctl disable --now nm-cloud-setup.service nm-cloud-setup.timer 2>/dev/null || true
cat > /etc/NetworkManager/conf.d/rke2-canal.conf <<'EOF'
[keyfile]
unmanaged-devices=interface-name:cali*;interface-name:flannel*
EOF
systemctl reload NetworkManager

# no proxy variables in an offline install
env | grep -i proxy
```

4.3 Base packages and installer

```
dnf install -y tar unzip curl jq lvm2 iptables nfs-utils   # from local repo

mkdir -p /opt/UiPathAutomationSuite/2025.10.2/installer
cd /opt/UiPathAutomationSuite/2025.10.2
unzip /tmp/as-installer.zip -d installer
chmod -R 755 installer
cd installer

lsblk -o NAME,SIZE,TYPE,FSTYPE,MOUNTPOINT      # identify disks
```

Also: root umask 0022; antivirus/EDR exclusions for /var/lib/rancher, /var/lib/kubelet and /opt/UiPathAutomationSuite; SELinux may stay enforcing.

4.4 Configure disks

Node-specific - see the disk command in each node's section (5 to 10).

4.5 cluster_config.json and prerequisite check

The file is created once on Server 1 (Section 5, step 5.4) and then copied unchanged to all other nodes. Once it is present on a node, run:

```
cd /opt/UiPathAutomationSuite/2025.10.2/installer
jq . cluster_config.json > /dev/null && echo 'JSON OK'
./bin/uipathctl prereq run cluster_config.json --versions versions/helm-charts.json
```

Fix every **FAIL** before continuing; review every WARN.

5. Server 1 (primary) - Phase A: infrastructure

Hostname	IP	Role	Holds
as-srv-01	10.0.0.11	control-plane + etcd (first node)	sf.tar.gz, sf-infra.tar.gz, /uipath, cluster_config.json (source)

5.1 Hostname

```
hostnamectl set-hostname as-srv-01
```

5.2 Common steps

Complete Section 4.1 - 4.3.

5.3 Configure disks (incl. bundle disk)

```
cd /opt/UiPathAutomationSuite/2025.10.2/installer
./configureUiPathDisks.sh --node-type server --install-type offline \
  --cluster-disk-name /dev/sdb \
  --etcd-disk-name    /dev/sdc \
  --data-disk-name    /dev/sdd \
  --bundle-disk-name  /dev/sde

# in-cluster Ceph object store (raw, unformatted disk)
./configureUiPathDisks.sh --ceph-raw-disk-name /dev/sdf

df -h /var/lib/rancher /var/lib/rancher/rke2/server/db /datadisk /uipath

mv /tmp/sf.tar.gz /tmp/sf-infra.tar.gz /uipath/
mkdir -p /uipath/tmp
sha256sum /uipath/*.tar.gz
```

5.4 Create cluster_config.json

Option A (recommended): run the wizard, choose *Multi-node HA-ready production* and *Offline*, enter FQDN, LB address, SQL, certificates and products. Stop after the config file is generated - the multi-node install continues manually.

```
cd /opt/UiPathAutomationSuite/2025.10.2/installer
./installUiPathAS.sh
```

Option B: write it manually. Key fields (schema per 2025.10 docs):

```
{
  "fqdn": "automationsuite.corp.local",
  "fixed_rke_address": "automationsuite-lb.corp.local",
  "admin_username": "admin",
  "admin_password": "<StrongPassword>",
  "rke_token": "<uuidgen output - same on all nodes>",
  "profile": "ha",
  "install_type": "offline",
  "telemetry_optout": false,
  "server_certificate": {
    "ca_cert_file": "/opt/certs/rootCA.crt",
    "tls_cert_file": "/opt/certs/server.crt",
    "tls_key_file": "/opt/certs/server.key" },
  "sql": { "server_url": "sql.corp.local", "port": "1433",
           "username": "uipathas", "password": "<pwd>", "create_db": true },
  "sql_connection_string_template": "...",
  "sql_connection_string_template_jdbc": "...",
  "sql_connection_string_template_odbc": "...",
  "sql_connection_string_template_sqlalchemy_pyodbc": "...",
  "infra": { "docker_registry": { "username": "admin", "password": "<pwd>" } },
  "external_object_storage": { "enabled": false },
  "orchestrator":  { "enabled": true },
  "action_center": { "enabled": true }
}
```

5.5 Copy the config to the other 5 nodes

```
for h in as-srv-02 as-srv-03 as-agt-01 as-agt-02 as-agt-03; do
  scp cluster_config.json root@$h:/opt/UiPathAutomationSuite/2025.10.2/installer/
done
```

5.6 Prerequisite check

```
cd /opt/UiPathAutomationSuite/2025.10.2/installer
./bin/uipathctl prereq run cluster_config.json --versions versions/helm-charts.json
```

5.7 Install RKE2 infrastructure (creates the cluster)

```
cd /opt/UiPathAutomationSuite/2025.10.2/installer
./bin/uipathctl rke2 install \
  -i ./cluster_config.json \
  -o ./output.json \
  --offline-bundle /uipath/sf-infra.tar.gz \
  --offline-tmp-folder /uipath/tmp \
  --install-type offline \
  --accept-license-agreement
```

5.8 Verify

```
export KUBECONFIG=/etc/rancher/rke2/rke2.yaml
export PATH=$PATH:/var/lib/rancher/rke2/bin
kubectl get nodes -o wide      # as-srv-01  Ready  control-plane,etcd,master
kubectl get pods -A | grep -Ev 'Running|Completed'
```

Add the two export lines to /root/.bashrc on Server 1 - all later kubectl commands assume them. **Next:** go to Server 2 (Section 6). Server 1 Phase B (Section 11) runs only after all 6 nodes are Ready.

Server 1 - Phase A checklist

Done	Task
[ ]	Hostname set, /etc/hosts, chrony synced
[ ]	Swap off, firewall, nm-cloud-setup disabled
[ ]	Installer unzipped to /opt/UiPathAutomationSuite/2025.10.2/installer
[ ]	Disks configured incl. /uipath bundle disk and Ceph raw disk
[ ]	sf.tar.gz + sf-infra.tar.gz in /uipath, checksums OK
[ ]	cluster_config.json created and copied to 5 nodes
[ ]	prereq run passed
[ ]	rke2 install completed, node Ready

6. Server 2 - join as server

Hostname	IP	Role	Holds
as-srv-02	10.0.0.12	control-plane + etcd	sf-infra.tar.gz, cluster_config.json

> **Start only when:** Server 1 is Ready (Section 5 completed).

6.1 Hostname

```
hostnamectl set-hostname as-srv-02
```

6.2 Common steps

Complete Section 4.1 - 4.3 (hosts file, chrony, swap, firewall, NetworkManager, packages, unzip installer).

6.3 Configure disks

```
cd /opt/UiPathAutomationSuite/2025.10.2/installer
./configureUiPathDisks.sh --node-type server --install-type offline \
  --cluster-disk-name /dev/sdb \
  --etcd-disk-name    /dev/sdc \
  --data-disk-name    /dev/sdd

# in-cluster Ceph object store (raw, unformatted disk)
./configureUiPathDisks.sh --ceph-raw-disk-name /dev/sdf

df -h /var/lib/rancher /var/lib/rancher/rke2/server/db /datadisk
```

No bundle disk on this node.

6.4 Stage the infrastructure bundle

```
mkdir -p /opt/UiPathAutomationSuite/bundle/tmp
mv /tmp/sf-infra.tar.gz /opt/UiPathAutomationSuite/bundle/
sha256sum /opt/UiPathAutomationSuite/bundle/sf-infra.tar.gz
df -h /opt        # needs enough free space to extract the bundle
```

6.5 Confirm config and run prerequisite check

```
cd /opt/UiPathAutomationSuite/2025.10.2/installer
ls -l cluster_config.json        # copied from Server 1 - do NOT edit
./bin/uipathctl prereq run cluster_config.json --versions versions/helm-charts.json

# LB reachability for join
curl -k https://automationsuite-lb.corp.local:9345/ping
timeout 3 bash -c '</dev/tcp/automationsuite-lb.corp.local/6443' && echo 6443 OK
```

6.6 Join the cluster as server

```
cd /opt/UiPathAutomationSuite/2025.10.2/installer
./bin/uipathctl rke2 install \
  -i ./cluster_config.json \
  -o ./output.json \
  -j server \
  --offline-bundle /opt/UiPathAutomationSuite/bundle/sf-infra.tar.gz \
  --offline-tmp-folder /opt/UiPathAutomationSuite/bundle/tmp \
  --install-type offline \
  --accept-license-agreement
```

6.7 Verify

```
# on as-srv-02
systemctl status rke2-server --no-pager
journalctl -u rke2-server -n 50 --no-pager

# on Server 1
kubectl get nodes -o wide          # as-srv-02 must be Ready
kubectl -n kube-system get pods -l component=etcd
```

Next: wait until as-srv-02 is Ready, then go to Server 3 (Section 7). Do not start Server 3 earlier.

Server 2 checklist

Done	Task
[ ]	Hostname set, /etc/hosts, chrony synced
[ ]	Swap off, firewall, nm-cloud-setup disabled
[ ]	Installer unzipped to /opt/UiPathAutomationSuite/2025.10.2/installer
[ ]	Disks configured (cluster + etcd + data + Ceph)
[ ]	sf-infra.tar.gz staged, checksum OK
[ ]	cluster_config.json identical to Server 1
[ ]	prereq run passed, LB 9345/6443 reachable
[ ]	rke2 install -j server completed
[ ]	as-srv-02 shows Ready in kubectl get nodes

7. Server 3 - join as server

Hostname	IP	Role	Holds
as-srv-03	10.0.0.13	control-plane + etcd	sf-infra.tar.gz, cluster_config.json

Start only when: Server 1 and Server 2 are both Ready. Never join two servers at the same time.

7.1 Hostname

```
hostnamectl set-hostname as-srv-03
```

7.2 Common steps

Complete Section 4.1 - 4.3 (hosts file, chrony, swap, firewall, NetworkManager, packages, unzip installer).

7.3 Configure disks

```
cd /opt/UiPathAutomationSuite/2025.10.2/installer
./configureUiPathDisks.sh --node-type server --install-type offline \
  --cluster-disk-name /dev/sdb \
  --etcd-disk-name    /dev/sdc \
  --data-disk-name    /dev/sdd

# in-cluster Ceph object store (raw, unformatted disk)
./configureUiPathDisks.sh --ceph-raw-disk-name /dev/sdf

df -h /var/lib/rancher /var/lib/rancher/rke2/server/db /datadisk
```

No bundle disk on this node.

7.4 Stage the infrastructure bundle

```
mkdir -p /opt/UiPathAutomationSuite/bundle/tmp
mv /tmp/sf-infra.tar.gz /opt/UiPathAutomationSuite/bundle/
sha256sum /opt/UiPathAutomationSuite/bundle/sf-infra.tar.gz
df -h /opt        # needs enough free space to extract the bundle
```

7.5 Confirm config and run prerequisite check

```
cd /opt/UiPathAutomationSuite/2025.10.2/installer
ls -l cluster_config.json        # copied from Server 1 - do NOT edit
./bin/uipathctl prereq run cluster_config.json --versions versions/helm-charts.json

# LB reachability for join
curl -k https://automationsuite-lb.corp.local:9345/ping
timeout 3 bash -c '</dev/tcp/automationsuite-lb.corp.local/6443' && echo 6443 OK
```

7.6 Join the cluster as server

```
cd /opt/UiPathAutomationSuite/2025.10.2/installer
./bin/uipathctl rke2 install \
  -i ./cluster_config.json \
  -o ./output.json \
  -j server \
  --offline-bundle /opt/UiPathAutomationSuite/bundle/sf-infra.tar.gz \
  --offline-tmp-folder /opt/UiPathAutomationSuite/bundle/tmp \
  --install-type offline \
  --accept-license-agreement
```

7.7 Verify

```
# on as-srv-03
systemctl status rke2-server --no-pager
journalctl -u rke2-server -n 50 --no-pager

# on Server 1
kubectl get nodes -o wide          # as-srv-03 must be Ready
kubectl -n kube-system get pods -l component=etcd
```

Next: all three servers Ready - continue with Agent 1, 2 and 3 (Sections 8-10, may run in parallel).

Server 3 checklist

Done	Task
[ ]	Hostname set, /etc/hosts, chrony synced
[ ]	Swap off, firewall, nm-cloud-setup disabled
[ ]	Installer unzipped to /opt/UiPathAutomationSuite/2025.10.2/installer
[ ]	Disks configured (cluster + etcd + data + Ceph)
[ ]	sf-infra.tar.gz staged, checksum OK
[ ]	cluster_config.json identical to Server 1
[ ]	prereq run passed, LB 9345/6443 reachable
[ ]	rke2 install -j server completed
[ ]	as-srv-03 shows Ready in kubectl get nodes

8. Agent 1 - join as agent

Hostname	IP	Role	Holds
as-agt-01	10.0.0.21	worker	sf-infra.tar.gz, cluster_config.json

Start only when: all three servers (as-srv-01/02/03) are Ready.

8.1 Hostname

```
hostnamectl set-hostname as-agt-01
```

8.2 Common steps

Complete Section 4.1 - 4.3 (hosts file, chrony, swap, firewall, NetworkManager, packages, unzip installer).

8.3 Configure disks

```
cd /opt/UiPathAutomationSuite/2025.10.2/installer
./configureUiPathDisks.sh --node-type agent --install-type offline \
  --cluster-disk-name /dev/sdb

df -h /var/lib/rancher
```

Agents have no etcd disk and no bundle disk.

8.4 Stage the infrastructure bundle

```
mkdir -p /opt/UiPathAutomationSuite/bundle/tmp
mv /tmp/sf-infra.tar.gz /opt/UiPathAutomationSuite/bundle/
sha256sum /opt/UiPathAutomationSuite/bundle/sf-infra.tar.gz
df -h /opt        # needs enough free space to extract the bundle
```

8.5 Confirm config and run prerequisite check

```
cd /opt/UiPathAutomationSuite/2025.10.2/installer
ls -l cluster_config.json        # copied from Server 1 - do NOT edit
./bin/uipathctl prereq run cluster_config.json --versions versions/helm-charts.json

# LB reachability for join
curl -k https://automationsuite-lb.corp.local:9345/ping
timeout 3 bash -c '</dev/tcp/automationsuite-lb.corp.local/6443' && echo 6443 OK
```

8.6 Join the cluster as agent

```
cd /opt/UiPathAutomationSuite/2025.10.2/installer
./bin/uipathctl rke2 install \
  -i ./cluster_config.json \
  -o ./output.json \
  -j agent \
  --offline-bundle /opt/UiPathAutomationSuite/bundle/sf-infra.tar.gz \
  --offline-tmp-folder /opt/UiPathAutomationSuite/bundle/tmp \
  --install-type offline \
  --accept-license-agreement
```

8.7 Verify

```
# on as-agt-01
systemctl status rke2-agent --no-pager
journalctl -u rke2-agent -n 50 --no-pager

# on Server 1
kubectl get nodes -o wide          # as-agt-01 must be Ready
```

Next: when Agent 1, 2 and 3 are all Ready, return to Server 1 Phase B (Section 11).

Agent 1 checklist

Done	Task
[ ]	Hostname set, /etc/hosts, chrony synced
[ ]	Swap off, firewall, nm-cloud-setup disabled
[ ]	Installer unzipped to /opt/UiPathAutomationSuite/2025.10.2/installer
[ ]	Disks configured (cluster disk)
[ ]	sf-infra.tar.gz staged, checksum OK
[ ]	cluster_config.json identical to Server 1
[ ]	prereq run passed, LB 9345/6443 reachable
[ ]	rke2 install -j agent completed
[ ]	as-agt-01 shows Ready in kubectl get nodes

9. Agent 2 - join as agent

Hostname	IP	Role	Holds
as-agt-02	10.0.0.22	worker	sf-infra.tar.gz, cluster_config.json

> **Start only when:** all three servers (as-srv-01/02/03) are Ready. Can run in parallel with Agent 1 and 3.

9.1 Hostname

```
hostnamectl set-hostname as-agt-02
```

9.2 Common steps

Complete Section 4.1 - 4.3 (hosts file, chrony, swap, firewall, NetworkManager, packages, unzip installer).

9.3 Configure disks

```
cd /opt/UiPathAutomationSuite/2025.10.2/installer
./configureUiPathDisks.sh --node-type agent --install-type offline \
  --cluster-disk-name /dev/sdb

df -h /var/lib/rancher
```

Agents have no etcd disk and no bundle disk.

9.4 Stage the infrastructure bundle

```
mkdir -p /opt/UiPathAutomationSuite/bundle/tmp
mv /tmp/sf-infra.tar.gz /opt/UiPathAutomationSuite/bundle/
sha256sum /opt/UiPathAutomationSuite/bundle/sf-infra.tar.gz
df -h /opt        # needs enough free space to extract the bundle
```

9.5 Confirm config and run prerequisite check

```
cd /opt/UiPathAutomationSuite/2025.10.2/installer
ls -l cluster_config.json        # copied from Server 1 - do NOT edit
./bin/uipathctl prereq run cluster_config.json --versions versions/helm-charts.json

# LB reachability for join
curl -k https://automationsuite-lb.corp.local:9345/ping
timeout 3 bash -c '</dev/tcp/automationsuite-lb.corp.local/6443' && echo 6443 OK
```

9.6 Join the cluster as agent

```
cd /opt/UiPathAutomationSuite/2025.10.2/installer
./bin/uipathctl rke2 install \
  -i ./cluster_config.json \
  -o ./output.json \
  -j agent \
  --offline-bundle /opt/UiPathAutomationSuite/bundle/sf-infra.tar.gz \
  --offline-tmp-folder /opt/UiPathAutomationSuite/bundle/tmp \
  --install-type offline \
  --accept-license-agreement
```

9.7 Verify

```
# on as-agt-02
systemctl status rke2-agent --no-pager
journalctl -u rke2-agent -n 50 --no-pager

# on Server 1
kubectl get nodes -o wide          # as-agt-02 must be Ready
```

> **Next:** when Agent 1, 2 and 3 are all Ready, return to Server 1 Phase B (Section 11).

Agent 2 checklist

Done	Task
[ ]	Hostname set, /etc/hosts, chrony synced
[ ]	Swap off, firewall, nm-cloud-setup disabled
[ ]	Installer unzipped to /opt/UiPathAutomationSuite/2025.10.2/installer
[ ]	Disks configured (cluster disk)
[ ]	sf-infra.tar.gz staged, checksum OK
[ ]	cluster_config.json identical to Server 1
[ ]	prereq run passed, LB 9345/6443 reachable
[ ]	rke2 install -j agent completed
[ ]	as-agt-02 shows Ready in kubectl get nodes

10. Agent 3 - join as agent

Hostname	IP	Role	Holds
as-agt-03	10.0.0.23	worker	sf-infra.tar.gz, cluster_config.json

Start only when: all three servers (as-srv-01/02/03) are Ready. Can run in parallel with Agent 1 and 2.

10.1 Hostname

```
hostnamectl set-hostname as-agt-03
```

10.2 Common steps

Complete Section 4.1 - 4.3 (hosts file, chrony, swap, firewall, NetworkManager, packages, unzip installer).

10.3 Configure disks

```
cd /opt/UiPathAutomationSuite/2025.10.2/installer
./configureUiPathDisks.sh --node-type agent --install-type offline \
  --cluster-disk-name /dev/sdb

df -h /var/lib/rancher
```

Agents have no etcd disk and no bundle disk.

10.4 Stage the infrastructure bundle

```
mkdir -p /opt/UiPathAutomationSuite/bundle/tmp
mv /tmp/sf-infra.tar.gz /opt/UiPathAutomationSuite/bundle/
sha256sum /opt/UiPathAutomationSuite/bundle/sf-infra.tar.gz
df -h /opt        # needs enough free space to extract the bundle
```

10.5 Confirm config and run prerequisite check

```
cd /opt/UiPathAutomationSuite/2025.10.2/installer
ls -l cluster_config.json        # copied from Server 1 - do NOT edit
./bin/uipathctl prereq run cluster_config.json --versions versions/helm-charts.json

# LB reachability for join
curl -k https://automationsuite-lb.corp.local:9345/ping
timeout 3 bash -c '</dev/tcp/automationsuite-lb.corp.local/6443' && echo 6443 OK
```

10.6 Join the cluster as agent

```
cd /opt/UiPathAutomationSuite/2025.10.2/installer
./bin/uipathctl rke2 install \
  -i ./cluster_config.json \
  -o ./output.json \
  -j agent \
  --offline-bundle /opt/UiPathAutomationSuite/bundle/sf-infra.tar.gz \
  --offline-tmp-folder /opt/UiPathAutomationSuite/bundle/tmp \
  --install-type offline \
  --accept-license-agreement
```

10.7 Verify

```
# on as-agt-03
systemctl status rke2-agent --no-pager
journalctl -u rke2-agent -n 50 --no-pager

# on Server 1
kubectl get nodes -o wide          # as-agt-03 must be Ready
```

Next: when Agent 1, 2 and 3 are all Ready, return to Server 1 Phase B (Section 11).

Agent 3 checklist

Done	Task
[ ]	Hostname set, /etc/hosts, chrony synced
[ ]	Swap off, firewall, nm-cloud-setup disabled
[ ]	Installer unzipped to /opt/UiPathAutomationSuite/2025.10.2/installer
[ ]	Disks configured (cluster disk)
[ ]	sf-infra.tar.gz staged, checksum OK
[ ]	cluster_config.json identical to Server 1
[ ]	prereq run passed, LB 9345/6443 reachable
[ ]	rke2 install -j agent completed
[ ]	as-agt-03 shows Ready in kubectl get nodes

11. Server 1 (primary) - Phase B: registry + UiPath services

> **Start only when:** all 6 nodes show Ready in `kubectl get nodes`.

11.1 Confirm all nodes

```
kubectl get nodes -o wide
# as-srv-01   Ready   control-plane,etcd,master
# as-srv-02   Ready   control-plane,etcd,master
# as-srv-03   Ready   control-plane,etcd,master
# as-agt-01   Ready   <none>
# as-agt-02   Ready   <none>
# as-agt-03   Ready   <none>
```

Optional: label/taint specialised agents (GPU for AI Center, Task Mining) per the docs before 11.3.

11.2 Hydrate the in-cluster registry

```
cd /opt/UiPathAutomationSuite/2025.10.2/installer
./bin/uipathctl rke2 registry upload \
  --offline-bundle /uipath/sf.tar.gz \
  --offline-tmp-folder /uipath/tmp
```

Repeat for each optional bundle (DU on-demand, CV, ...). Expect 1-2+ hours depending on disk speed.

11.3 Install Fabric and UiPath products

```
cd /opt/UiPathAutomationSuite/2025.10.2/installer
./bin/uipathctl prereq run cluster_config.json --versions versions/helm-charts.json

./bin/uipathctl manifest apply cluster_config.json --versions versions/helm-charts.json
```

11.4 Monitor (second shell)

```
watch -n 20 "kubectl get pods -A | grep -Ev 'Running|Completed'"
kubectl -n argocd get applications      # all Synced / Healthy
```

Server 1 - Phase B checklist

Done	Task
[ ]	All 6 nodes Ready
[ ]	registry upload of sf.tar.gz completed
[ ]	Optional bundles uploaded (if any)
[ ]	prereq run passed against live cluster
[ ]	manifest apply completed
[ ]	All ArgoCD applications Synced / Healthy

12. Validation and post-installation (Server 1)

```
cd /opt/UiPathAutomationSuite/2025.10.2/installer
./bin/uipathctl health check
./bin/uipathctl health bundle    # support bundle if anything fails
```

Portal	URL	Credentials
Host portal	https://automationsuite.corp.local/ (Host)	admin / admin_password from config
Default organization	https://automationsuite.corp.local/	orgadmin / same password (change at first login)
ArgoCD	https://alm.automationsuite.corp.local	see docs (uipathctl config argocd)
Monitoring	https://monitoring.automationsuite.corp.local	see docs

Post-install tasks

• Activate the license (offline activation in Host portal -> Licenses).

• Replace self-signed certificates: `./bin/uipathctl config tls-certificates update --tls-cert server.crt --tls-key server.key --ca-cert ca.crt`

• Configure backup (etcd snapshots + NFS backup target).

• Configure SMTP, identity provider (AD / Entra ID / SAML) and alert receivers.

• Keep the /uipath bundle disk - it is needed for adding nodes later.

• Store cluster_config.json securely (contains passwords) and back it up off-cluster.

• Connect Robots / Studio to https://automationsuite.corp.local/<org>/<tenant>/orchestrator_

13. Troubleshooting

Symptom	Where	Check
Server node does not join	Server 2 / 3	journalctl -u rke2-server -f; LB 9345/6443 reachable; same rke_token in config
Agent node does not join	Agent 1-3	journalctl -u rke2-agent -f; LB 9345 reachable; time in sync
etcd unhealthy	Servers	etcd disk latency (fio), clock skew (chronyc tracking)
Pods in ImagePullBackOff	Server 1	Registry upload incomplete - repeat Section 11.2
ArgoCD app Degraded	Server 1	kubectl -n argocd get app <name> -o yaml, then pod logs
SQL errors	Any	Connection strings, port 1433 from every node, dbcreator role
DNS failures inside pods	Any	/etc/resolv.conf has only reachable nameservers; custom_dns_resolver
Anything else	Server 1	uipathctl health check / health bundle -> UiPath support ticket

14. Master sign-off checklist

Milestone	Srv 1	Srv 2	Srv 3	Agt 1	Agt 2	Agt 3
OS prepared (Section 4)	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
Installer + sf-infra copied	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
sf.tar.gz in /uipath	[ ]	n/a	n/a	n/a	n/a	n/a
Disks configured	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
cluster_config.json present	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
prereq run passed	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
rke2 install done / Ready	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
registry upload	[ ]	n/a	n/a	n/a	n/a	n/a

Milestone	Srv 1	Srv 2	Srv 3	Agt 1	Agt 2	Agt 3
manifest apply	[ ]	n/a	n/a	n/a	n/a	n/a
health check green	[ ]	n/a	n/a	n/a	n/a	n/a

Sign-off	Name	Date	Signature
Installed by			
Verified by			